



Python Programming Fundamentals

String Methods

Programming Fundamentals Team

SETU

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Outline

1. Strings are Objects
2. Case and Whitespace Methods
3. Search and Replace
4. Split and Join
5. f-Strings
6. String Formatting Specifiers
7. String Methods in Practice



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1. Strings are Objects

In Python, **everything is an object** — including strings. Objects have **methods** (functions attached to them) called with dot notation.

```
name = "hello world"
print(type(name))           # <class 'str'>
```

```
# Calling a method on a string object
print(name.upper())         # HELLO WORLD
print(name.capitalize())    # Hello world
print(name.title())         # Hello World
```



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In Python, **everything is an object** — including strings. Objects have **methods** (functions attached to them) called with dot notation.

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print(type(name))           # <class 'str'>
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```
# Calling a method on a string object
print(name.upper())         # HELLO WORLD
print(name.capitalize())    # Hello world
print(name.title())         # Hello World
```

Strings are immutable — methods return **new** strings, they don't modify the original:

```
greeting = "hello"
upper_greeting = greeting.upper()
print(greeting)             # hello (unchanged)
print(upper_greeting)       # HELLO
```



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2. Case and Whitespace Methods

```
text = " Hello, World! "
```

```
print(text.upper())      # " HELLO, WORLD! "
print(text.lower())      # " hello, world! "
print(text.strip())      # "Hello, World!"      removes leading/trailing
spaces
print(text.lstrip())      # "Hello, World! "      left strip
print(text.rstrip())      # " Hello, World!"      right strip
```

```
name = "  alice  "
clean_name = name.strip().title()
print(clean_name)        # "Alice"
```



2. Case and Whitespace Methods



2. Case and Whitespace Methods

```
text = "  Hello, World!  "
```

```
print(text.upper())      # "  HELLO, WORLD!  "
print(text.lower())      # "  hello, world!  "
print(text.strip())      # "Hello, World!"      removes leading/trailing
spaces
print(text.lstrip())      # "Hello, World!  "    left strip
print(text.rstrip())      # "  Hello, World!"    right strip
```

```
name = "  alice  "
clean_name = name.strip().title()
print(clean_name)        # "Alice"

# Checking methods – return True/False
print("PYTHON".isupper()) # True
print("python".islower()) # True
```



2. Case and Whitespace Methods

```
print("Python123".isalnum())    # True (letters and digits)
print(" ".isspace())           # True
```



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3. Search and Replace

```
sentence = "The quick brown fox jumps over the lazy dog"
```

```
# Finding substrings
```

```
print(sentence.find("fox"))          # 16   (index of first match)
print(sentence.find("cat"))          # -1   (not found)
print(sentence.count("the"))         # 1    (case-sensitive)
print(sentence.count("the", 0, -1, ))
```

```
# Checking start/end
```

```
print(sentence.startswith("The"))    # True
print(sentence.endswith("dog"))      # True
print(sentence.startswith("the"))    # False (case-sensitive)
```



3. Search and Replace



3. Search and Replace

```
sentence = "The quick brown fox jumps over the lazy dog"
```

```
# Finding substrings
```

```
print(sentence.find("fox"))          # 16   (index of first match)
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```

```
# Checking start/end
```

```
print(sentence.startswith("The"))    # True
print(sentence.endswith("dog"))      # True
print(sentence.startswith("the"))    # False (case-sensitive)
```

```
# Replacing
```

```
new_sentence = sentence.replace("fox", "cat")
print(new_sentence)
```



3. Search and Replace

```
# "The quick brown cat jumps over the lazy dog"

# Replace only first N occurrences
text = "aaa bbb aaa ccc aaa"
print(text.replace("aaa", "XXX", 2))
# "XXX bbb XXX ccc aaa"
```



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4. Split and Join

split() — break a string into a list of parts:

```
csv_line = "Alice,25,Dublin,Engineer"
parts = csv_line.split(",")
print(parts)    # ['Alice', '25', 'Dublin', 'Engineer']
```

```
sentence = "the quick brown fox"
words = sentence.split()    # splits on any whitespace
print(words)    # ['the', 'quick', 'brown', 'fox']
```



4. Split and Join



4. Split and Join

split() — break a string into a list of parts:

```
csv_line = "Alice,25,Dublin,Engineer"
parts = csv_line.split(",")
print(parts)      # ['Alice', '25', 'Dublin', 'Engineer']
```

```
sentence = "the quick brown fox"
words = sentence.split()      # splits on any whitespace
print(words)      # ['the', 'quick', 'brown', 'fox']
```

join() — glue a list of strings together:

```
words = ["Python", "is", "great"]
print(" ".join(words))      # Python is great
print("-".join(words))      # Python-is-great
print("".join(words))      # Pythonisgreat
```



4. Split and Join

```
# Practical: rebuild a CSV line
data = ["Bob", "30", "Cork", "Teacher"]
csv = ",".join(data)
print(csv)    # Bob,30,Cork,Teacher
```



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5. f-Strings

f-strings (formatted string literals) embed expressions directly in strings.

```
name = "Alice"  
age = 25  
score = 87.654
```

```
# Basic embedding
```

```
print(f"Name: {name}, Age: {age}")
```

```
# Expressions inside {}
```

```
print(f"Next year I'll be {age + 1}")
```

```
# Number formatting
```

```
print(f"Score: {score:.2f}")          # 87.65
```

```
print(f"Score: {score:.0f}")          # 88
```

```
print(f"Score: {score:8.2f}")          # "    87.65" (width 8)
```



5. f-Strings

Alignment

```
print(f"{'Left':<10}|")
```

```
print(f"{'Right':>10}|")
```

```
print(f"{'Centre':^10}|")
```

```
# Left      |
```

```
#      Right |
```

```
#   Centre   |
```



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6. String Formatting Specifiers

Specifier	Meaning	Example
:.2f	2 decimal places	3.14
:d	Integer	42
:,	Thousands separator	1,000,000
:%	Percentage	75.00%
:>10	Right-align width 10	hello
:<10	Left-align width 10	hello
:^10	Centre width 10	hello

```
price = 1234567.89
print(f"Price: €{price:,.2f}")      # Price: €1,234,567.89
```



6. String Formatting Specifiers

```
ratio = 0.756
print(f"Pass rate: {ratio:.1%}")    # Pass rate: 75.6%

items = [("Milk", 1.29), ("Bread", 2.49), ("Eggs", 3.99)]
for item, cost in items:
    print(f"    {item:<10} €{cost:>6.2f}")
```



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7. String Methods in Practice

```
def clean_and_validate_email(email):
    """Clean and validate an email address."""
    cleaned = email.strip().lower()

    if "@" not in cleaned:
        return None, "Missing @ symbol"
    if "." not in cleaned.split("@")[1]:
        return None, "Invalid domain"

    return cleaned, "Valid"

def title_case_name(name):
    """Clean and format a person's name."""
    return " ".join(word.capitalize() for word in name.strip().split())

# Test
```



7. String Methods in Practice

```
tests = [" Alice Smith ", "BOB JONES", "carol ann murphy"]
for name in tests:
    print(title_case_name(name))
```



Thanks for Watching - Any questions?

